

The Law States

"For every action, there is an equal and opposite reaction."

1 Introduction

Sir Isaac Newton published his three laws of motion in 1687 in his famous work *Philosophiæ Naturalis Principia Mathematica*. The **Third Law** describes the fundamental nature of forces: they always occur in **pairs**. Whenever one object exerts a force on a second object, the second object simultaneously exerts a force of equal magnitude but opposite direction on the first object.

2 The Mathematical Form

$$\mathbf{F}_{A \rightarrow B} = -\mathbf{F}_{B \rightarrow A}$$

The force exerted by object A on object B is equal in magnitude and opposite in direction to the force exerted by object B on object A.

The negative sign indicates that the two forces point in **opposite directions**. Importantly, these two forces act on **different objects** — which is why they never cancel each other out.

3 Visualizing the Law

A B F (A on B) F (B on A)

Figure 1: Action–reaction force pair. Equal in magnitude, opposite in direction, acting on different objects.

4 Everyday Examples



Rocket Propulsion

A rocket pushes hot gases downward (action); the gases push the rocket upward (reaction), lifting it into space.



Walking

Your foot pushes backward on the ground (action); the ground pushes your foot forward (reaction), moving you ahead.



Swimming

A swimmer pushes water backward (action); the water pushes the swimmer forward (reaction).



Gun Recoil

A gun pushes the bullet forward (action); the bullet pushes the gun backward (reaction), causing recoil.



Balloon Release

Air rushes out of a balloon in one direction (action); the balloon flies off in the opposite direction (reaction).



Rowing a Boat

Oars push the water backward (action); the water pushes the boat forward (reaction).

5 Properties of Action–Reaction Pairs

Property	Description
Equal Magnitude	Both forces have exactly the same strength.
Opposite Direction	The forces point in exactly opposite directions.
Different Objects	Each force acts on a different body — they never cancel out.
Simultaneous	Both forces occur at the same instant; there is no delay.
Same Type	Both forces are of the same nature (both gravitational, both contact, etc.).

Common Misconception

"If the forces are equal and opposite, why don't they cancel out?"

Action and reaction forces act on *different objects*. Forces can only cancel when they act on the *same object*. When you push a wall, your force acts on the wall, while the wall's reaction force acts on *you* — two different bodies, so no cancellation occurs.

Key Takeaways

- Forces always exist in pairs — a single isolated force is impossible.
- Action and reaction are equal in magnitude and opposite in direction.
- The paired forces act on two different objects, so they never cancel.

- Both forces act simultaneously — neither one comes "first."
- This law explains propulsion: rockets, walking, swimming, and flight.

6 Worked Example

Problem: A book of mass 2 kg rests on a table. The book presses down on the table with a force of 19.6 N. What force does the table exert on the book?

Solution: By Newton's Third Law, the table exerts an equal and opposite force of **19.6 N upward** on the book (called the *normal force*). This is why the book remains at rest — the upward push of the table balances the downward pull of gravity acting on the book.

"To every action there is always opposed an equal reaction." — Sir Isaac Newton, 1687